

# CORRECTED SHOULD-COST ANALYSIS — Bosch IPB2.0 ECU PCBA

Commodity: PCBA (populated assembly) · Currency: GBP · Region: China · Annual volume: 150,000 · Operations: 4

Re-costed from eight board photographs after the uploaded tool report costed this board as bare *pcb\_fab* with zero operations. Bottom-up, fully traceable.

| TOTAL SHOULD-COST | MATERIAL | PROCESS | LABOUR | OVERHEAD | MARGIN |
|-------------------|----------|---------|--------|----------|--------|
| £184.40           | 72.4%    | 10.7%   | 1.1%   | 7.6%     | 7.4%   |

## §1 — Key Assumptions

Board 170×110 mm (187 cm²) · 6 layers · FR4 High-Tg 150C · 2 oz outer / 1 oz inner copper · ENIG · AEC-Q Grade 2 / PPAP  
755 SMT placements · 3 BGA · 2 press-fit connectors · double-sided assembly · assembly quality grade AEC-Q Grade 1 (brake-by-wire)  
Overhead 9% of factory base · Margin 8% of subtotal · Yield 98.5% first-pass · NRE £46,000 assembly + £9,500 fab, both amortised over 150,000.  
**Board size and layer count are scaled from the photographs, not measured.** A caliper measurement and a layer count from a cross-section would remove the largest single uncertainty in this estimate.

## §2 — 8-Bucket Cost Breakdown

| Cost Bucket                              | Amount (GBP)   | % of Total    |
|------------------------------------------|----------------|---------------|
| 1. Raw Material (bare PCB + BOM + yield) | £133.56        | 72.4%         |
| 2. Process (Machine)                     | £19.65         | 10.7%         |
| 3. Direct Labour                         | £1.94          | 1.1%          |
| 4. Tooling (amortised)                   | £0.31          | 0.2%          |
| 5. Packaging                             | £0.73          | 0.4%          |
| 6. Logistics                             | £0.57          | 0.3%          |
| <b>FACTORY COST</b>                      | <b>£156.75</b> | <b>85.0%</b>  |
| 7. Overhead (SG&A)                       | £13.99         | 7.6%          |
| <b>SUBTOTAL</b>                          | <b>£170.74</b> | <b>92.6%</b>  |
| 8. Supplier Margin                       | £13.66         | 7.4%          |
| <b>TOTAL SHOULD-COST</b>                 | <b>£184.40</b> | <b>100.0%</b> |

## §3 — Bill of Materials (the uploaded tool report contained no BOM at all)

| Ref              | Qty | Description                                | Unit £        | Ext £ | Source                                         |
|------------------|-----|--------------------------------------------|---------------|-------|------------------------------------------------|
| ✓ U-MCU          | 1   | Renesas R7F702300B RH850 automotive MCU    | 15.000        | 15.00 | marking R7F702300B FAB-A-C 2308 JAPAN          |
| ✓ U-ASIC1        | 1   | Bosch 40342/01 motor-control ASIC          | 10.000        | 10.00 | marking 40342 /01 2324 VC252NVN                |
| ✓ U-ASIC2        | 1   | Bosch 40341/01 ASIC                        | 7.000         | 7.00  | marking 40341 /01 A2192F4A                     |
| ✓ U-ASIC3        | 1   | Bosch 2302701 ASIC                         | 4.000         | 4.00  | marking 2302701                                |
| ✓ U-SBC          | 1   | ST L9369 multi-channel driver / SBC        | 5.000         | 5.00  | marking L9369 VC DTE                           |
| ~ U-BGA2         | 1   | Secondary safety MCU / ASIC (BGA)          | 8.000         | 8.00  | package visible, marking not legible           |
| ~ U-MEM          | 1   | External memory (BGA)                      | 3.000         | 3.00  | package visible, marking not legible           |
| ✓ Q1-6           | 6   | Q142E power stage / dual half-bridge       | 3.000         | 18.00 | marking Q142E BABA W31C ×6                     |
| ✓ U-CAN          | 2   | NXP TJA1463A CAN-FD SIC transceiver        | 1.200         | 2.40  | marking TJA1463A                               |
| ✓ U-71H          | 3   | Bosch 71H740 driver / power-path           | 1.500         | 4.50  | marking 71H740 Prm 2321 C3                     |
| ✓ U-76E2         | 1   | Bosch 76E240 device                        | 1.800         | 1.80  | marking 76E240 PNW 2328 B6                     |
| ✓ U-76E8         | 1   | Bosch 76E840 device                        | 1.800         | 1.80  | marking 76E840 PEm 2319 A2                     |
| ✓ U-7S1R         | 2   | Bosch 7S1R540H power device                | 1.600         | 3.20  | marking 7S1R540H Prt 2313D6                    |
| ✓ C-BULK         | 6   | Polymer alu cap 150 µF 35 V (351 150 EJV)  | 0.400         | 2.40  | marking 351 150 EJV ×6                         |
| ~ L-PWR          | 8   | Shielded power inductor 0.47–10 µH         | 0.600         | 4.80  | 0.47µH/10µH legible; count from both sides     |
| ~ R/C-SMD        | 620 | MLCC + thick-film resistors, AEC-Q200      | 0.009         | 5.58  | density count across both populated sides      |
| ~ D/Q-SML        | 90  | Small-signal semis (SOT/SOD/QFN)           | 0.090         | 8.10  | density count; individual markings not legible |
| ~ Y1             | 1   | Crystal / oscillator                       | 0.450         | 0.45  | package visible near MCU                       |
| ~ J1-2           | 2   | Automotive press-fit header (main + motor) | 2.500         | 5.00  | through-hole field, bottom-left                |
| ✓ TIM            | 8   | Thermal interface pad (power stages)       | 0.120         | 0.96  | pink pads visible on underside                 |
| <b>BOM TOTAL</b> |     |                                            | <b>110.99</b> |       |                                                |

✓ = package marking legible in a photograph (13 of 20 lines, 69% of BOM value). ~ = count or price inferred from board density and class medians. Component prices are China buy, 150k/yr, and carry roughly ±25% until quoted — that band is the second-largest uncertainty after board size.

## §4 — Operations Detail (the uploaded tool report had zero operations)

| Operation                           | Machine             | Rate £/hr | Cycle (s) | OEE  | Machine £    | Labour £    | Op Total £   |
|-------------------------------------|---------------------|-----------|-----------|------|--------------|-------------|--------------|
| SMT Placement & Reflow              | smt-high-speed-line | 78.38     | 303       | 82%  | 8.06         | 0.55        | 8.60         |
| TH Insertion & Hand Soldering       | bench-assembly      | 0.36      | 24        | 100% | 0.00         | 0.04        | 0.05         |
| BGA X-Ray Inspection (100% offline) | xray-bga-inspection | 47.63     | 475       | 90%  | 6.99         | 0.86        | 7.84         |
| ICT / Functional Test               | ict-automotive      | 58.39     | 270       | 95%  | 4.61         | 0.49        | 5.10         |
| <b>TOTAL</b>                        |                     |           |           |      | <b>19.65</b> | <b>1.94</b> | <b>21.59</b> |

Bare board is bought in at £21.64 each (costed separately through the PCB-fab module: £18.06 board material and fab processing + tooling, packaging, logistics, overhead and margin) and enters this assembly as material, which is how a contract assembler actually buys it.

## §5 — Uncertainty and Sensitivity

| Driver                   | Basis                                          | Effect on total                       |
|--------------------------|------------------------------------------------|---------------------------------------|
| Board size / layer count | scaled from photos, not measured               | ±8% (±£15)                            |
| Component prices         | class medians, not quoted                      | ±25% of BOM → ±£28                    |
| BGA X-ray method         | headline uses 100% offline X-ray (475 s/board) | inline AXI at 45 s → £176.75 (–£7.66) |
| Assembly complexity      | 'high' — >300 parts, BGAs, double-sided        | ±10% of SMT → ±£1                     |
| Overhead / margin        | 9% / 8% assumed                                | ±1pt each → ±£3                       |

**Working range £165 – £215 per board.** At 150,000/yr, 100% offline X-ray would need 3.3 dedicated machines; inline AXI is the industry norm at this volume and is shown above as the realistic lower case. The headline keeps the conservative default — an over-estimate is a negotiating position, an under-estimate is a signed mistake. Inline AXI is now a selectable strategy on the PCBA form, so this case can be quoted directly rather than reconstructed by hand.

## §6 — Gap Analysis vs the Uploaded Tool Report

| # | Gap                            | What the tool report did                                                                                                         | Corrected here                                                                                                                                         | Severity        |
|---|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| 1 | <b>Wrong commodity</b>         | Costed as <b>pcb_fab</b> (bare board only), with one material line passed straight through.                                      | A populated assembly must be costed as <b>pcba</b> . Bare board is one input to it, not the whole part.                                                | <b>Critical</b> |
| 2 | <b>Zero operations</b>         | Process £0.00, Labour £0.00, "Operations: 0" — no placement, reflow, inspection or test.                                         | 4 operations, £21.59 of conversion cost. A 755-placement safety ECU cannot assemble itself.                                                            | <b>Critical</b> |
| 3 | <b>Volume ignored</b>          | Report shows "Annual volume: —"; tooling amortised over <b>5,000</b> parts.                                                      | 150,000/yr applied. Tooling falls from £4.52-equivalent to £0.31/part.                                                                                 | <b>Critical</b> |
| 4 | <b>No BOM</b>                  | One line: mat-virtual £82.43. No component lines, so the largest cost bucket is unauditable.                                     | 20 BOM lines, £110.99, each flagged confirmed or estimated (\$3).                                                                                      | <b>Critical</b> |
| 5 | <b>Wrong-commodity advice</b>  | §11 recommended "HPDC → lost foam; forging → closed-die" and a material-utilisation lever — casting and forging advice on a PCB. | Fixed in the engine: material-dominance advice is now gated to the commodity, and a pass-through material bucket no longer offers a utilisation lever. | <b>Major</b>    |
| 6 | <b>Carbon = 0.00</b>           | Net weight 0.000 kg → 0.00 kgCO2e for a 500 g electronics assembly.                                                              | Carbon must come from board area, copper mass and component class, not net weight.                                                                     | <b>Major</b>    |
| 7 | <b>Confidence not enforced</b> | Published a headline while self-reporting "Low confidence, 1 data point auditable".                                              | A costing with zero operations should be blocked, not rendered.                                                                                        | <b>Major</b>    |
| 8 | <b>Currency</b>                | Reported in CNY with an FX note, when GBP was asked for.                                                                         | Reported in GBP throughout.                                                                                                                            | <b>Minor</b>    |

Four engine defects found during this exercise have been fixed at source and pinned with tests: double-sided assembly was charging every component placement twice; BGA X-ray had no model other than 100% offline inspection; the PCB forms inherited mechanical-part packaging and freight defaults; and optimisation advice was not gated to the commodity. Gaps 1–4 and 6–8 above are how the report was *driven*, not engine faults — the board was costed under the wrong commodity from the start.

## §7 — Net Effect

|                         | Total per board | Material       | Process + Labour | Basis                                      |
|-------------------------|-----------------|----------------|------------------|--------------------------------------------|
| Uploaded tool report    | £98.06          | £82.43         | £0.00            | 0 operations · volume not applied · no BOM |
| Corrected (this report) | £184.40         | £133.56        | £21.59           | 4 operations · 150,000/yr · 20-line BOM    |
| <b>Difference</b>       | <b>+£86.34</b>  | <b>+£51.13</b> | <b>+£21.59</b>   | <b>tool under-states by 47%</b>            |

The gap is not a disagreement about component prices — the two BOM values are within a quarter of each other. It is that the uploaded report contained **no assembly cost whatsoever** and never applied the 150,000/yr volume. Taking £98/board into a supplier meeting for this assembly would have put the floor price roughly half where it belongs.